

Rajesh Kanikkaiselvam

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EXPERIENCE SUMMARY

IoT Systems & Edge Computing Engineer specializing in embedded systems, Linux-based platforms, industrial networking, and cloud-connected cyber-physical systems. Experienced in developing real-time sensing, control, and data acquisition solutions using C/C++, Python, MQTT, Docker, ESP32, Raspberry Pi, and cloud IoT technologies. Volunteer Research Assistant at Illinois Institute of Technology contributing to intelligent monitoring and automation systems. Passionate about building secure, scalable, and distributed IoT architectures that bridge embedded devices, edge computing, and cloud infrastructure.

PROFESSIONAL EXPERIENCE

Cloud Systems & AI integration Engineer, Bluecurrent Tech (Remote) | *Chicago, IL* *Jan 2026 – Mar 2026*

- Audited cloud-to-mobile communication protocols, mapped end-to-end data flows across ISP/firewall/API layers, implemented targeted fixes, and conducted AI model error analysis to reduce bias and improve accuracy of tree scanning systems.
- Authored API developer documentation covering endpoint specs, JSON schemas, OAuth/JWT authentication, and mobile-cloud sync workflows while refining AI datasets through iterative curation, standardizing integration and accelerating team onboarding.

Junior Design Requirements Analyst, Vectra Automation, Inc. | *Chennai, India* *Feb 2023 – July 2023*

- Analyzed and translated customer and engineering requirements into system-level specifications for industrial automation projects.
- Collaborated with cross-functional teams to support design reviews, technical documentation, and requirement traceability throughout the product development lifecycle.
- Contributed to the development of reliable automation solutions by ensuring alignment between functional requirements, engineering designs, and project objectives.

TECHNICAL SKILLS

- **IoT & Cyber-Physical Systems:** Internet of Things (IoT), Cyber-Physical Systems (CPS), Device-to-Cloud Architectures, Edge-to-Cloud Integration, Distributed Systems, Industrial IoT (IIoT), Smart Monitoring Systems.
- **Programming:** Python, C, C++, SQL, Bash.
- **Embedded Systems & Edge Computing:** ESP32, Raspberry Pi, Embedded Linux, GPIO, UART, SPI, I2C, Sensor Integration, Real-Time Data Acquisition, Edge Computing.
- **Networking & Communication:** TCP/IP, UDP, MQTT, HTTP/REST APIs, WebSockets, Wi-Fi, Bluetooth Low Energy (BLE), Network Troubleshooting.
- **Cloud & DevOps:** Docker, Kubernetes, Git, GitHub, CI/CD, Linux Administration, Cloud-Native Systems.
- **Cybersecurity:** Computer Network Security, IoT Security Fundamentals, Authentication & Authorization, OAuth/JWT, Secure Communication Protocols.
- **AI & Intelligent Systems:** Machine Learning, Deep Learning, Computer Vision, OpenCV, TensorFlow, PyTorch, Edge AI, Time-Series Analysis.
- **Wireless & Communication Systems:** 5G Networks, Wireless Communication Systems, Network Architecture, Modern Internet Technologies.
- **Data Science & Analytics:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Tableau.
- **Databases:** MySQL, PostgreSQL, NoSQL.

SELECTED PROJECTS

Deep Reinforcement Learning-Based Real-Time Temperature Control for Thermoelectric Systems *Fall-2026*
Python | Q-Learning | PID Control | Embedded Systems | Cyber-Physical Systems

- **Developed** a hybrid Q-Learning and PID control system for real-time temperature regulation of a thermoelectric (Peltier) platform, enabling autonomous controller optimization on physical hardware
- **Built** a hardware-in-the-loop cyber-physical system integrating Arduino, DS18B20 sensors, relay control, and Modbus power management, executing 300+ automated reinforcement learning training sessions without simulation pre-training.
- **Engineered** a scalable experimental framework featuring automated data logging, fault recovery, and reproducible training workflows for reinforcement learning-based control system research.

AI Thermal Imaging for Predictive Tree Health Assessment *Jan 2026 – Mar 2026*
Computer Vision | Thermal Imaging | Deep Learning | Predictive Analytics |

- **Developed** an AI-driven thermal imaging system for predictive tree health assessment, computer vision and thermal analysis to detect early physiological stress, disease, and structural defects before visible symptoms appeared.

EDUCATION

Illinois Institute of Technology, Chicago | *MAS – Computer Engineering in Internet of Things* *F2025 - S2027*

Coursework: Machine Learning; Deep Learning; Big Data; Probabilistic Graphic; Statistical Analysis; Computer vision

Anna University, Chennai | *Bachelors of Engineering in Mechatronics* *2019 - 2023*